CORRELATION & REGRESSION

FA51 - Prof. Robbins
January 31, 2026

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Research Question

In the population of astrologers in the Clear Thinking in Astrology Challenge, can the number of years of experience predict the correct matches of natal charts while controlling for belief in astrology?

Background and Motivation

People believe that their natal chart, created with exact birth date, time, and location, contains much information about their character and life at a rate at least moderately better than random chance.

I wanted to test whether experience in astrology is actually associated with better performance, while controlling for belief by excluding those who do not believe in astrology. This allows me to examine whether experience itself adds explanatory power beyond belief alone.

Sample Dataset

Individuals who voluntarily attended the Clearer Thinking Astrology Challenge and reported positive belief in astrology.

Note: Belief in astrology was measured on a scale from −3 (total skeptic) to +3 (total believer). For the regression analysis, the sample was restricted to participants with belief scores greater than 0, indicating belief in astrology. This restriction reflects the theoretical assumption that meaningful engagement with astrological interpretation occurs primarily among believers (Carlson, 1985) and helps reduce multicollinearity, as belief and years of experience were correlated in the full sample ($r \approx 0.4$).

Population

The population consists of Individuals who would participate in an online astrology skill challenge like the Clearer Thinking Astrology Challenge under similar recruitment conditions (newsletter subscribers, social media outreach, etc) and believe in their ability to outperform chance.

Variables

	Predictor 1 | Independent: Experience years in astrolgy	Response | Dependent: Correct maches of natal charts
Type	Discrete Quantitative	Discrete Quantitative
Explanantion	This variable represents the number of years of experience in astrology. Participants reported their experience on a discrete scale ranging from 0 to 5 years, making this a discrete quantitative variable. It is used to examine the association between astrological experience and the number of correct matches.	This variable represents the number of correct matches between a person's natal chart and their personal information, out of a total of 12. The number of correct matches ranges from 0 to 12, making this a discrete quantitative variable. It is used to examine differences in performance based on astrologers' level of experience.

Summary Statistics for Predictor Variable

Sample size	109
Mean	2.17
Median	2
Mode	2
Range	5
SD	1.2

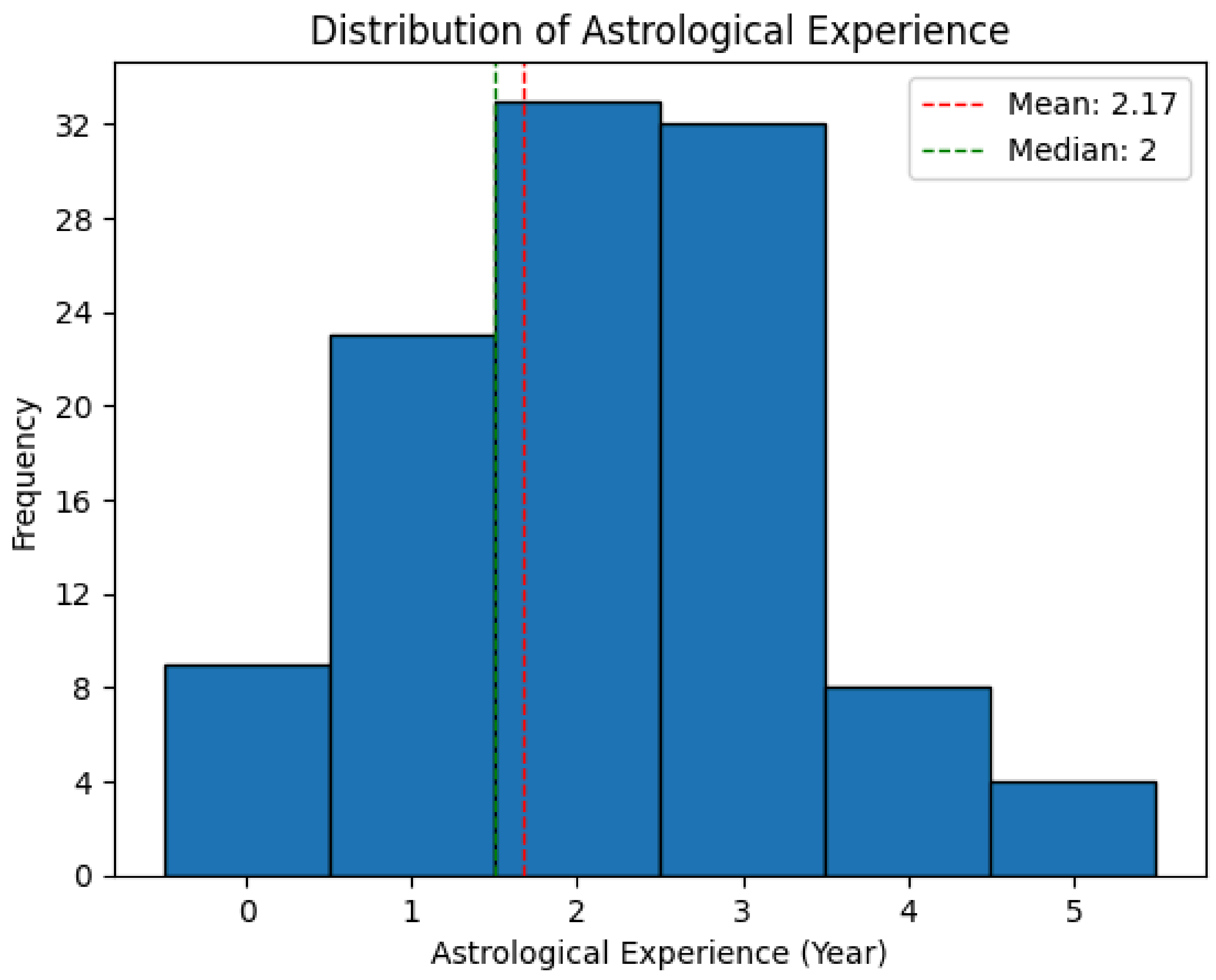


Figure 1. Histogram of years of astrological experience among participants, with dashed lines indicating the mean (2.17 years) and median (2 years).

Summary Statistics for Response Variable

Sample size	109
Mean	2.5
Median	2
Mode	2
Range	5
SD	1.39

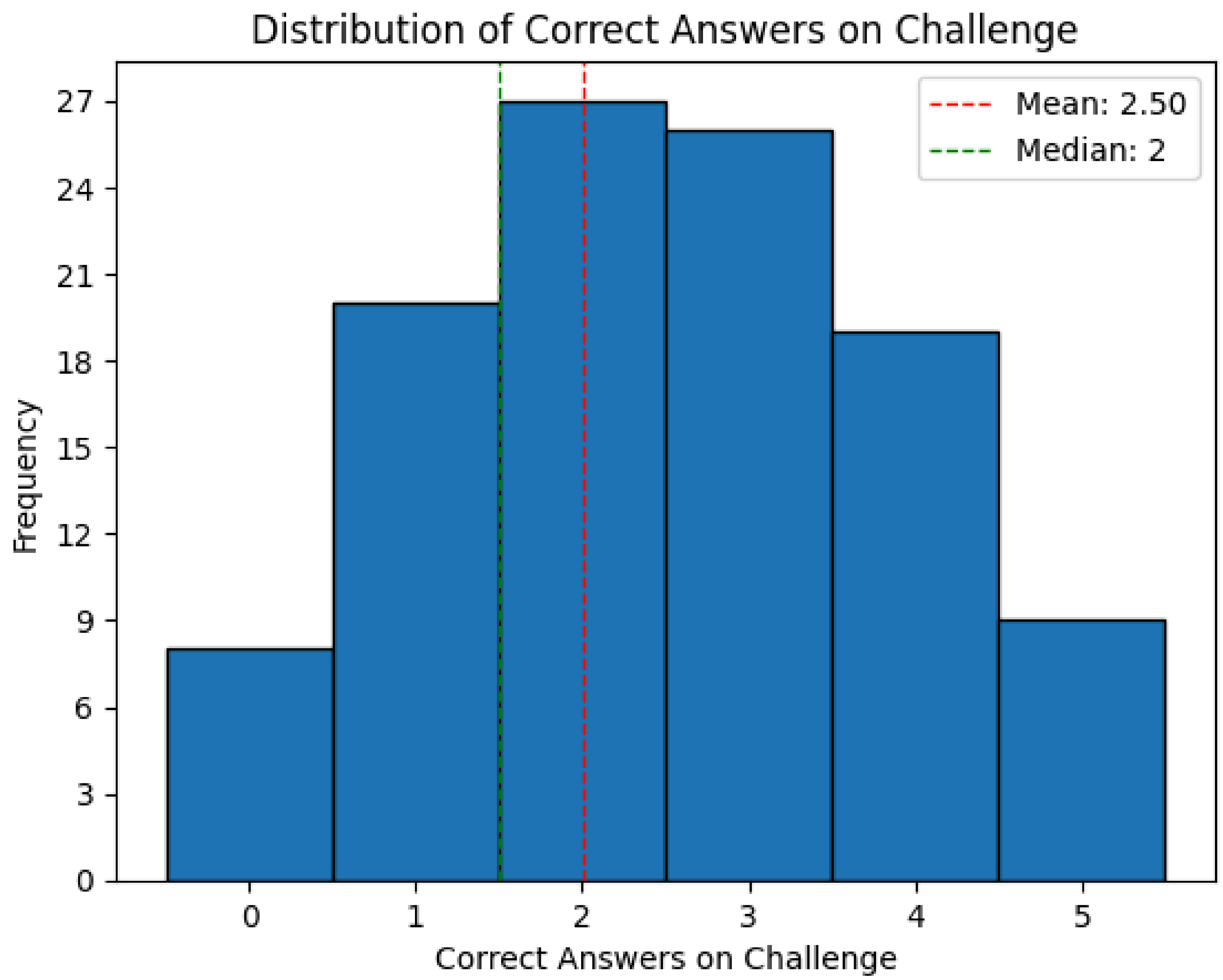


Figure 2. Histogram of number of correct answers among participants, with dashed lines indicating the mean (2.5 questions) and median (2 questions).

Correlation

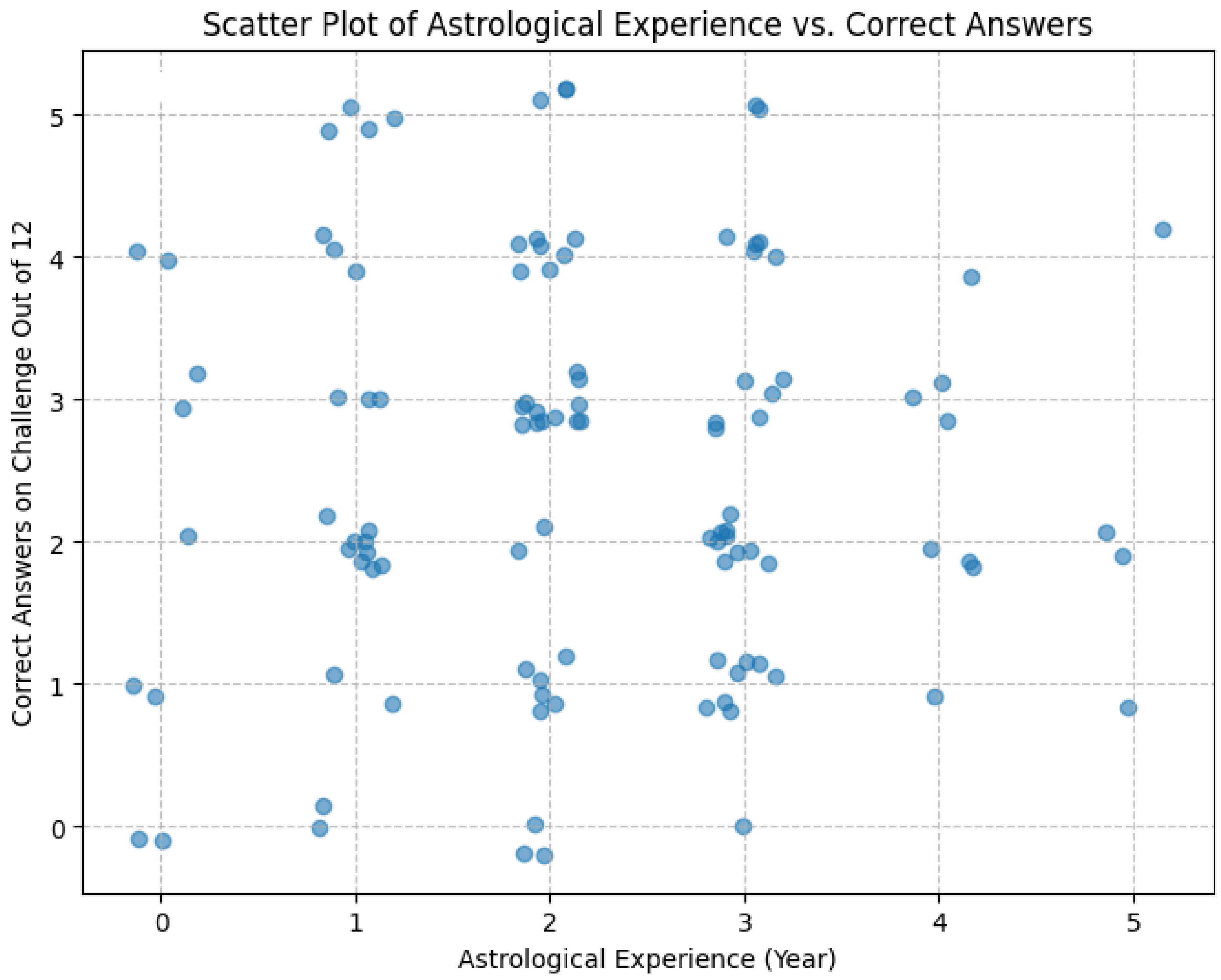


Figure 3. Scatter plot showing the relationship between years of astrological experience and number of correct matches.

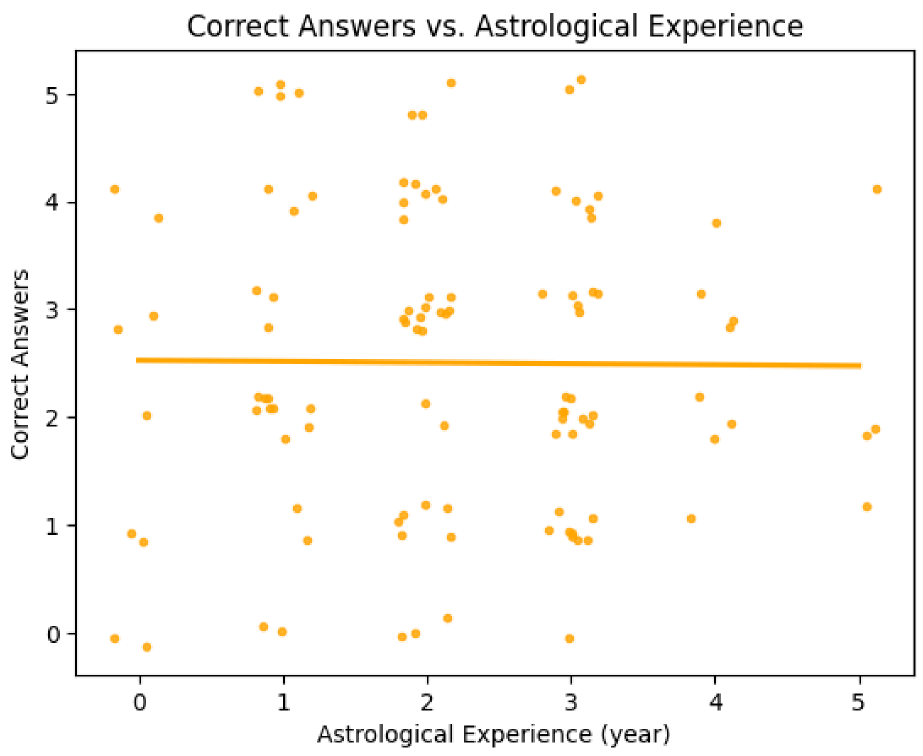
Note: Jitter added to improve the visualization (Waskom & the seaborn development team, n.d.)

r = -0.009

There is a very small negative correlation between the astrological experience and correct answers

- **L**inearity
- **I**ndependece
- **N**ormally distributed residuals
- **E**qual variance

Model Validation - 1



Linearity

The scatter plot and the fitted regression line suggest a linear trend. While the individual points are somewhat dispersed around the line, the overall pattern of the means follows a straight-line relationship.

Figure 4. Scatter plot showing the linearity in the model.

Independence

The data were collected randomly by the Clearer Thinking Organization, which promoted the survey through various channels to reach a diverse range of astrologers

Model Validation - 2

- Linearity
- Independece
- Normally distributed residuals
- Equal variance

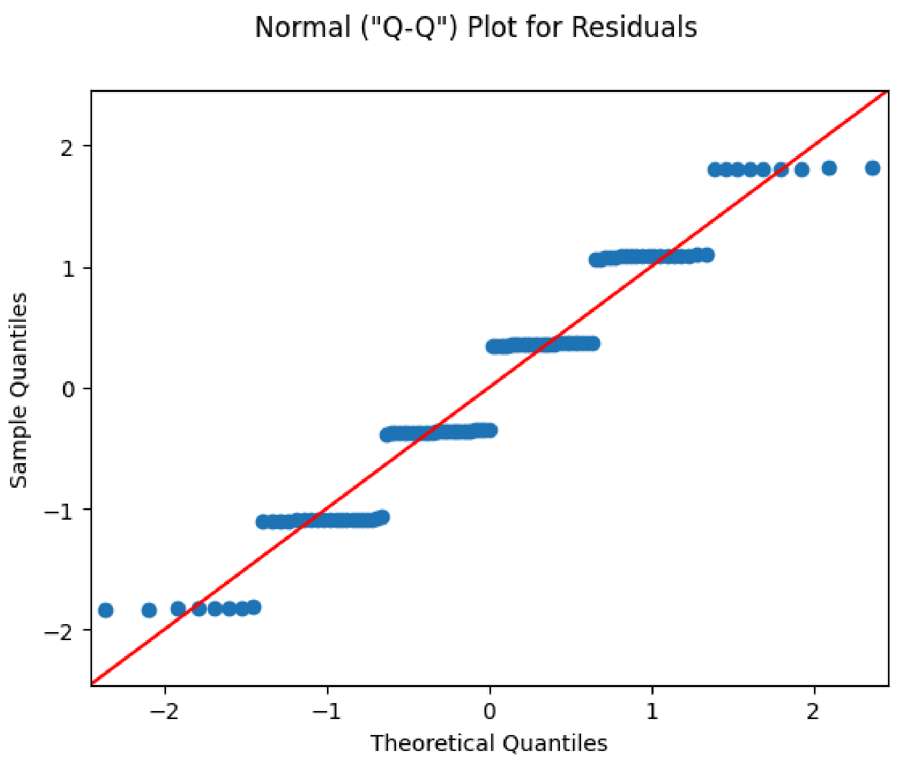


Figure 5. QQ plot showing the normality of residuals

Normally distributed residuals

Even though the data doesn't look completely normal, we can still see the familiar normal patterns throughout the graph.

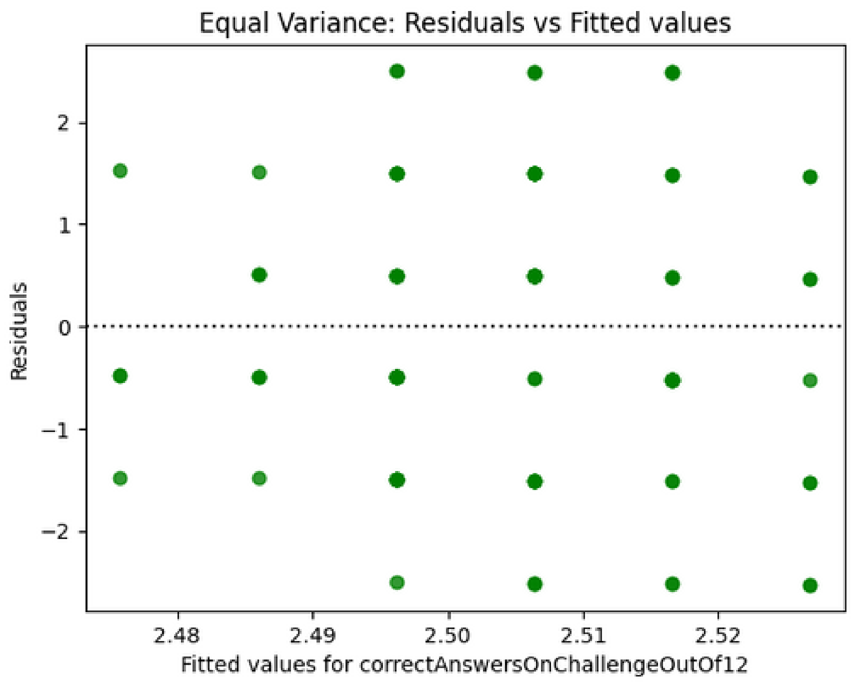
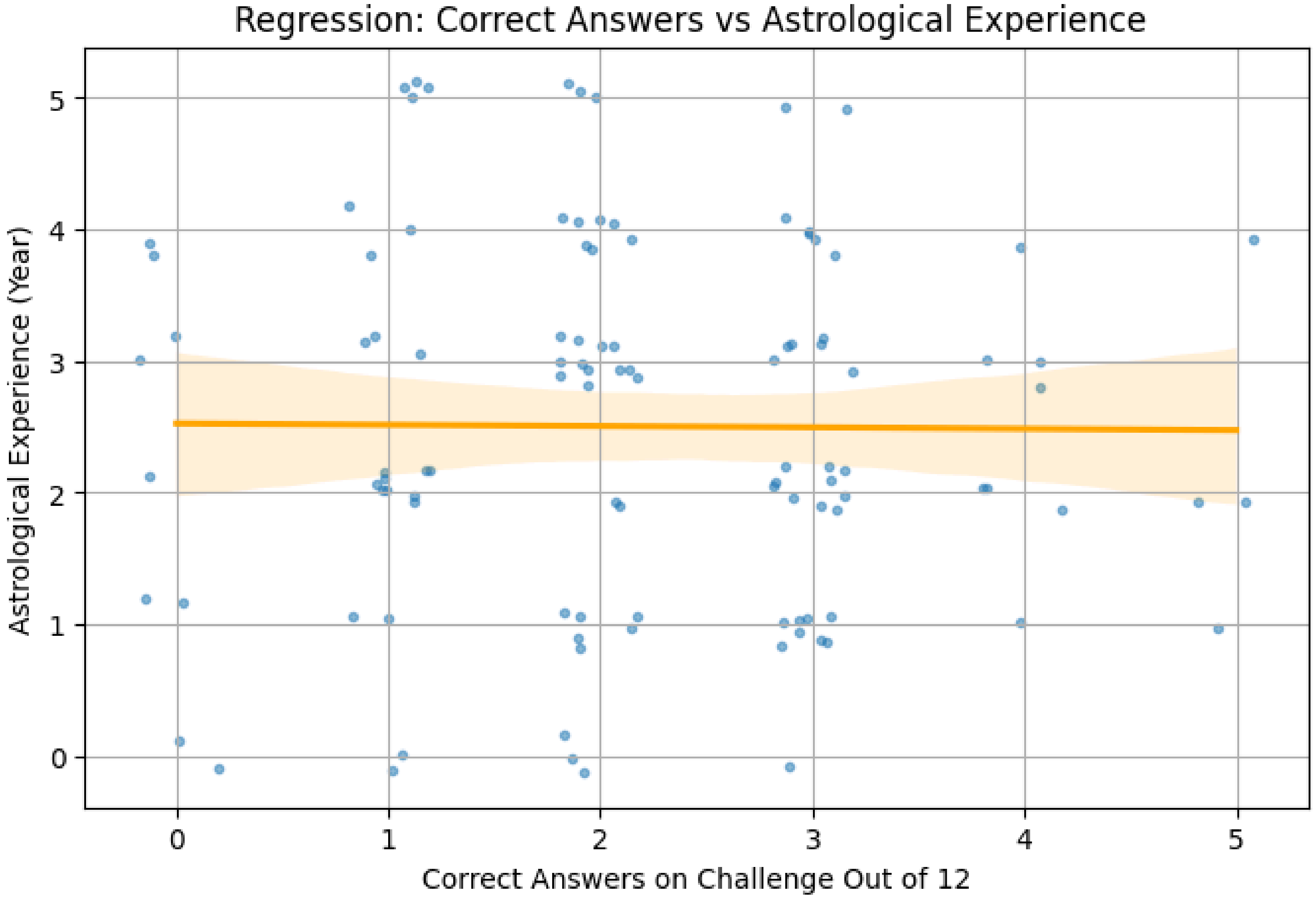


Figure 6. Scatter plot showing the scattering of data points

Equal Variance (Homoscedasticity)

Data points are scattered around almost the entire graph, not clustered in one area.

Regression



$R^2 = 0.0001$

0% of the variation in correct answers can be explained by astrological experience

Regression Equitation

$\hat{y} = 2.527 - 0.0102x$

Correct Answers = (intercept) + (coef of astrological experience) * (astrological experience)

Figure 7. Regression plot illustrating the relationship between astrological experience and correct matches. The fitted regression line shows no linear association.

Note: Jitter added to improve the visualization (Waskom & the seaborn development team, n.d.)

Significance

Null Hypothesis (H0): The number of years of experience and belief in astrology has no linear relationship with the number of correct natal chart matches, holding belief in astrology constant ($\beta1=0$).

Alternative Hypothesis (H1): The number of years of experience and belief in astrology has a linear relationship with the number of correct natal chart matches, holding belief in astrology constant ($\beta1\neq 0$).

SE: 0.1115
t-value= -0.091
p-value= 0.927
Significance Level (α)= 0.05

Two-tailed Test Result:
We fail to reject null hypothesis.

Confidence Interval

95% CI for slope:
[-0.2313, 0.2109]

The 95% confidence interval for the slope of the regression line is [-0.2313, 0.2109]. Since this interval contains zero, it confirms that we cannot rule out the possibility that the true population slope is zero. This interval is relatively wide compared to the slope itself, reflecting the high degree of uncertainty in using years of experience to predict matching accuracy within this specific group of believers.

Conclusion

The results demonstrate that among astrology believers, years of experience have virtually no predictive power over matching accuracy ($r=-0.009$, $R^2=0.0001$). Inductively, this suggests that, for the broader practitioner population, accumulated years of study do not translate into higher success rates on standardized natal chart-matching tasks. A major limitation of this model is that it does not account for an astrologer's experience or education, focusing only on duration, which appears to be an insufficient metric for skill. Additionally, the sample size is too small to make a general conclusion.

HC Explanantions 1

#correlation: I calculated Pearson's r to quantify the linear relationship between years of experience and matching accuracy. The result ($r = -0.0088$) indicates a negligible negative correlation, suggesting that these variables are essentially independent. My scatterplot confirms this lack of association, as the data points show no trend or clustering around the regression line.

#regression: I derived the linear model to predict matching accuracy based on experience. The coefficient of determination reveals that experience explains only 0.01% of the variation in scores, leaving 99.99% to be explained by other factors. I validated the model's assumptions using a Q-Q plot and a residual plot, which confirmed that while the residuals are normally distributed, the model has virtually no predictive power.

#dataviz: The initial scatter plot was unclear because the variables are discrete quantitative, leading to multiple observations overlapping at the same points and making interpretation difficult. Based on feedback from my PCW in Session 5, I added jitter to plots to add small amounts of noise, allowing the accumulation of data points at each value to be more clearly observed. These visualizations effectively communicate the "null" nature of the result, showing that the regression line is nearly flat and the data is widely dispersed.

HC Explanantions 2

#significance: I conducted a hypothesis test where the null hypothesis stated that the slope of experience is zero. The resulting p-value of 0.9273 is far above the significance threshold of 0.05, meaning the data is highly compatible with the null hypothesis. Therefore, I fail to reject the null hypothesis and conclude that there is no statistically significant relationship between experience and accuracy in this population.

#confidenceintervals: I computed a 95% confidence interval for the slope, resulting in [-0.2313, 0.2109]. Because this interval includes zero, it captures the possibility that the true population relationship is nonexistent. This range of both positive and negative values highlights the high degree of uncertainty in the estimate and reinforces the conclusion that experience is not a reliable predictor.

#professionalism: All resources for topics not covered but used in my assignment have been added, properly cited with annotations, and are formatted in accordance with APA style. Any usage of the AI tool is documented with links and explanations.

#variable: The predictor and response variables are Years of Astrological Experience and Correct Matches, both analyzed as quantitative discrete data. I applied jitter to the scatter plots to resolve overplotting and more accurately visualize the density of these discrete values. To control for confounding and mitigate multicollinearity, the sample was restricted to "Believers," ensuring the model evaluates the specific population where astrological experience is theorized to have the most impact.

Acknowledgement

I used the Gemini 3 (February 2026) integrated into the Colab notebook, where I wrote the code and created my visuals, and asked questions for clarification. When I encountered errors or the code did not run correctly, I asked the AI to help revise the code, which saved time compared to debugging it myself. Also, I used to get help implementing a new concept we didn't learn in class because revising the existing code is more complex and creates many bugs.

The code and the chat (accessible by clicking the blue dot in the bottom middle) can be found in the link below:

https://colab.research.google.com/drive/1ohw-f_BuArmWg8GWplb0XMGF8sPle100?usp=sharing

Bibliography

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